Extracting Signal from the Noise

The Unified Risk Intelligence Platform (URIP) Blueprint

Enterprise Orchestration Architecture | Confidential

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The Illusion of Security in a Siloed Environment

14 Siloed Consoles

Disconnected dashboards across VM, EASM, Cloud, OT, and VAPT. No unified view of the attack surface.

Manual Excel Registers

Hundreds of hours wasted copying and pasting data. No automated SLA tracking, no accountability, and immediate data decay.

Resource Drain

Low-impact theoretical vulnerabilities are addressed while active zero-days wait in the queue.

Blind Spots

Shadow IT remains invisible, and accepted risks artificially inflate board-level security reports.

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The CVSS Paradox: Why Spreadsheets Fail

The Decoy

Risk: CVSS 9.8 (Critical)

Context: No active exploitation globally. Low-tier lab asset.

Legacy Excel Result: Sent to the top of the queue. Team wastes 48 hours patching.

URIP Action: SSVC = TRACK. (Deprioritised).

The Weapon

Risk: CVSS 7.2 (High)

Context: CISA-confirmed active weaponisation. EPSS 95%. Running on a Tier 1 ERP system targeted by APT28.

Legacy Excel Result: Buried on row 412. Front door remains open.

URIP Action: Composite Score 9.3. SSVC = ACT NOW. Auto-assigned with a 72-hour SLA.

ACT NOW

A standard spreadsheet sorts by raw numbers and sends your team after the wrong threat.
URIP ranks by real-world attacker activity.

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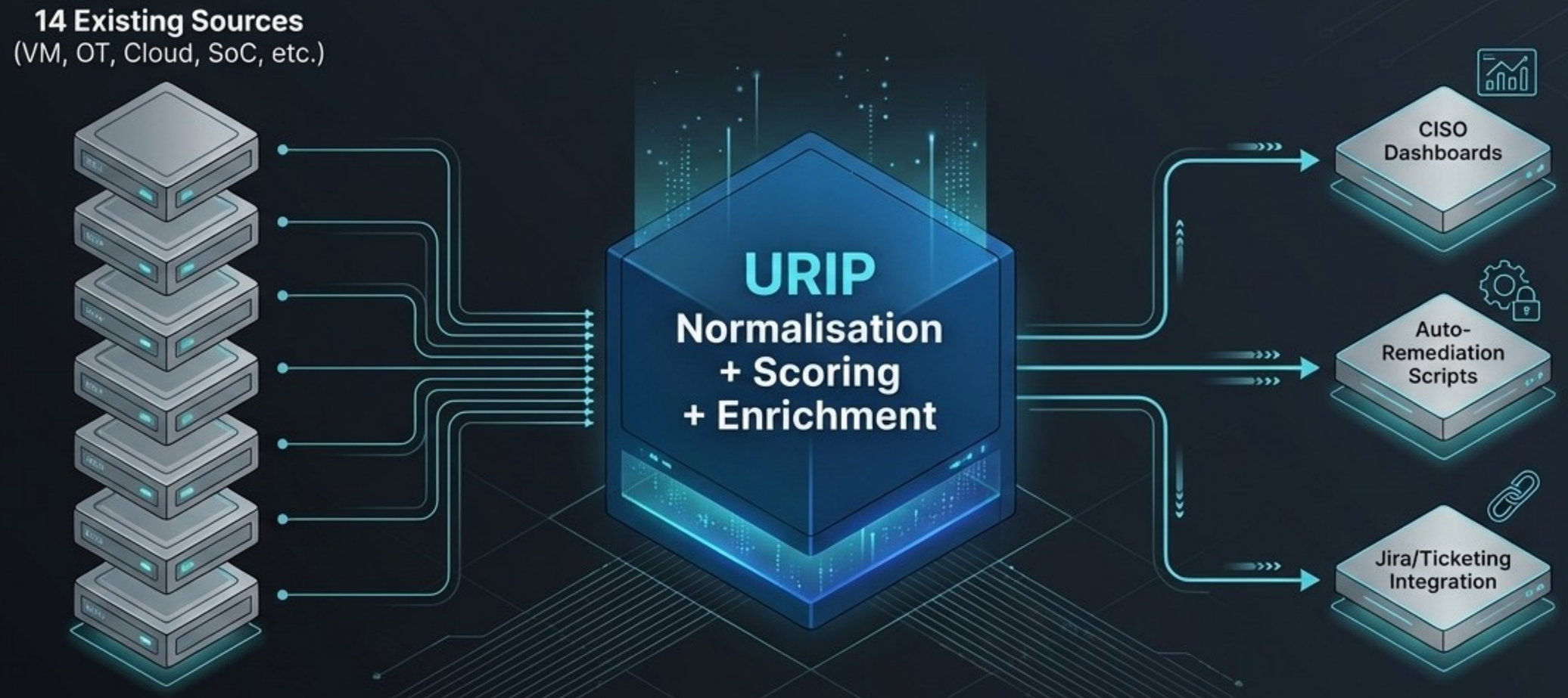
An Intelligence Overlay, Not a Replacement

URIP acts as a vendor-agnostic orchestration layer. It sits above your existing security stack to aggregate, prioritise, and automate.



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The End-to-End Orchestration Pipeline

1. DETECT

Pull live API feeds from 14 disparate risk sources.

2. AGGREGATE

Deduplicate via asset fingerprinting and normalise to a 0-10 scale.

3. PRIORITISE

Apply EPSS, KEV, Asset Tier, and APT intelligence.

4. ASSIGN

Auto-route to product owners and trigger SLA timers.

5. REMEDIATE

Attach mitigation steps and push auto-execution scripts.

6. REPORT

Generate CISO dashboards and clean board reports.

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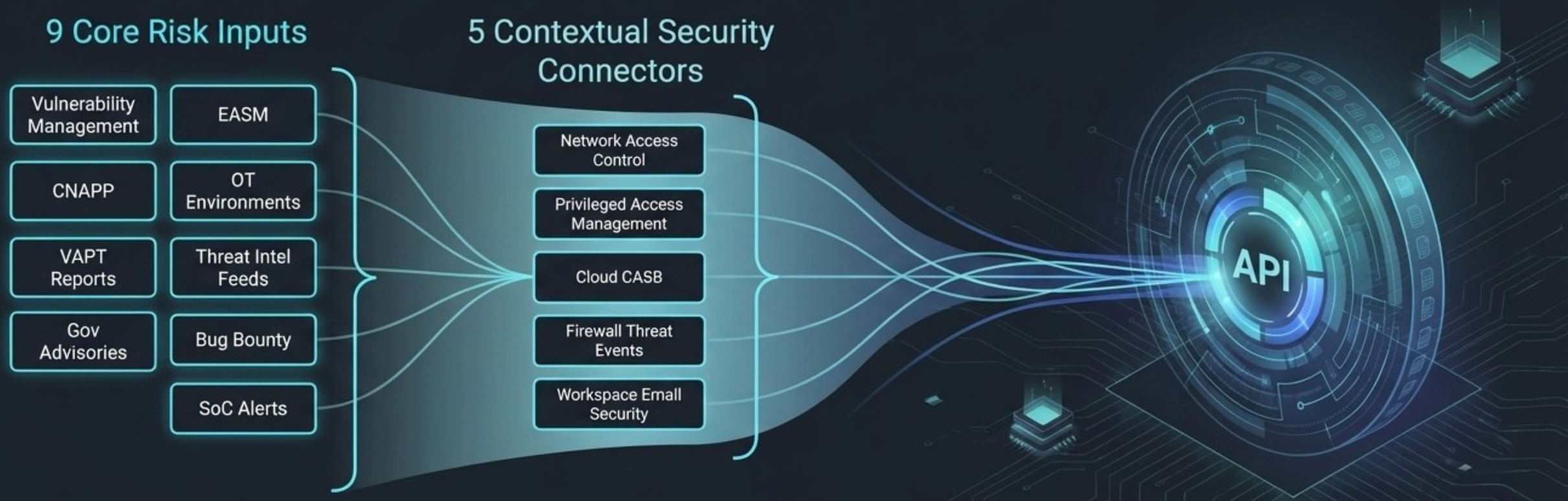
Ingestion: Taming 14 Disparate Risk Sources



Takeaway: All feeds connect via read-only APIs or webhooks. Zero manual data entry.

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Aggregation: The Deduplication Engine

Input A

VM Tool reports "Log4j High" on IP 192.168.1.50.

Input B

External VAPT PDF reports "CVE-2021-44228 Critical" on SAP-PRD-01.

The Engine: URIP Asset Fingerprinting (Matching MAC + Hostname + IP).

Output: A single, merged finding. The highest composite score is retained, both sources are attributed in the audit trail, and duplicate work is eliminated.

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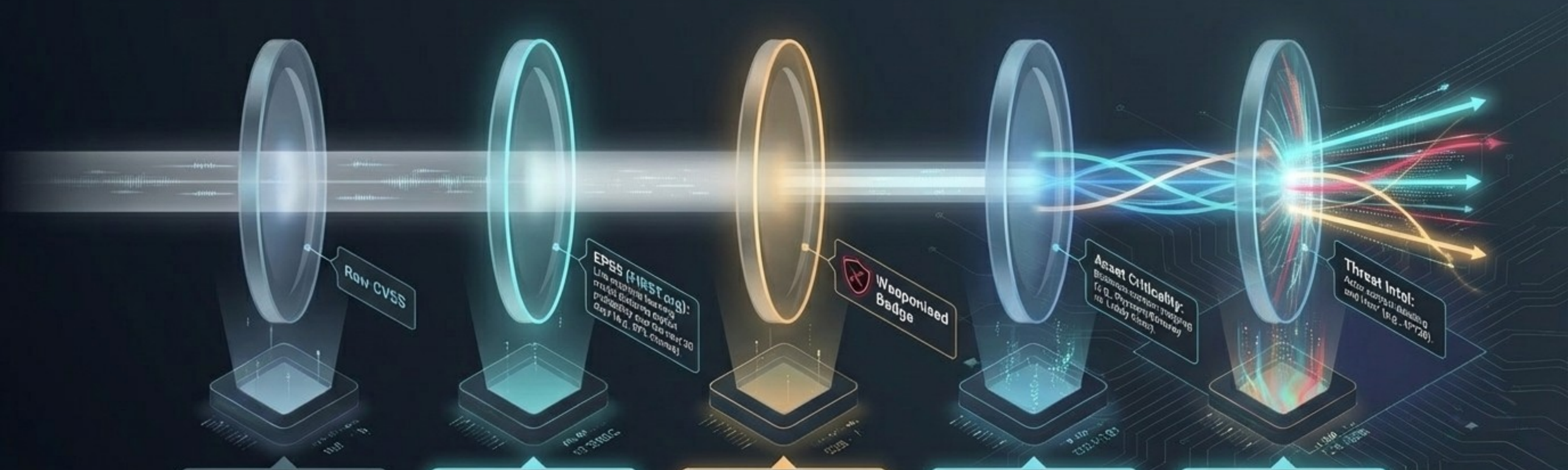
Prioritisation: The 5-Layer Intelligence Filter



Raw CVSS: Theoretical base severity (e.g., CVSS 10.0).

EPSS (FIRST.org): Live machine learning model dictating exploit probability over the next 30 days (e.g., 97% chance).

CISA KEV: Federal catalog confirming active exploitation in the wild.

Asset Criticality: Business context mapping (e.g., Payment Gateway vs. Lobby Kiosk).

Threat Intel: Actor context detailing "Who, Where, and How" (e.g., APT28).

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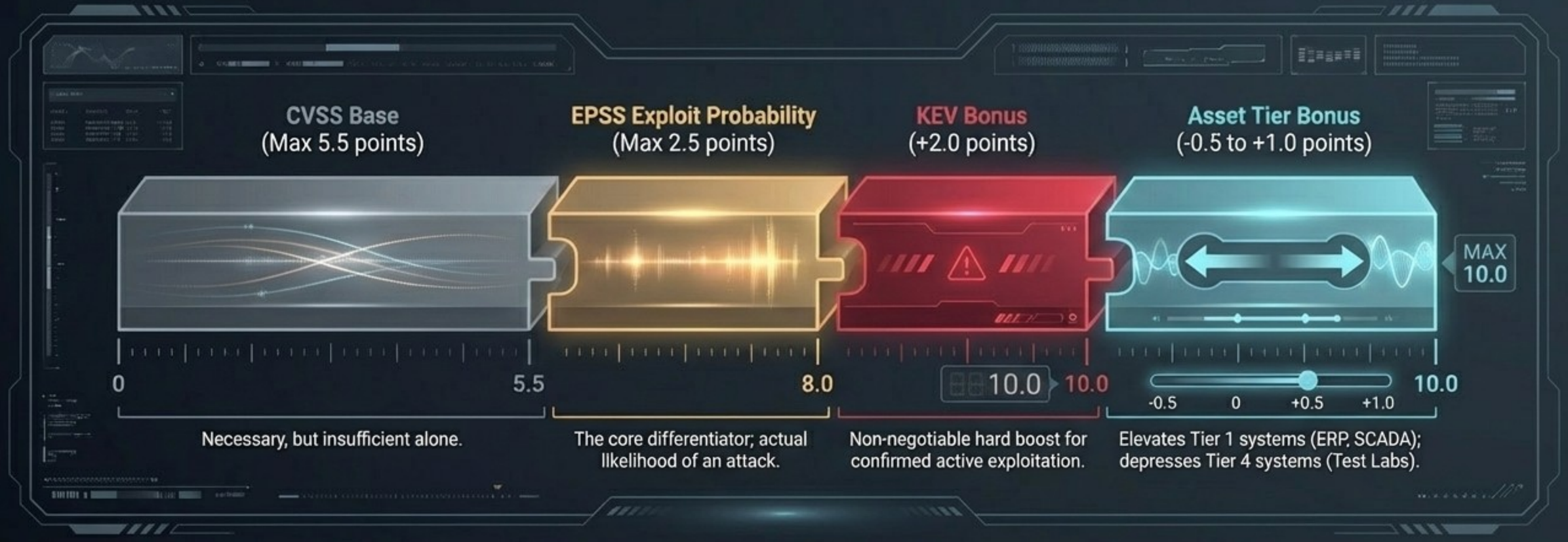
Anatomy of a Composite Score



Logic Note: The formula is strictly additive, not multiplicative. This prevents erratic mathematical amplification and guarantees that every point of a risk score can be clearly explained to an auditor or the board.

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The Engine in Action: 4 Real-World Scenarios

	CVE	Base CVSS	EPSS	CISA KEV	Asset Tier	URIP Composite	Action
Maximum Threat	PAN-OS (CVE-2024-3400)	10.0	0.97	YES	Tier 1 (+1.0)	10.0	ACT NOW
The Sneaky Threat	Cisco IOS (CVE-2023-20198)	7.2	0.95	YES	Tier 1 (+1.0)	9.3	ACT NOW
The Noisy Decoy	PuTTY (CVE-2024-31497)	5.9	0.02	NO	Tier 4 (-0.5)	2.8	TRACK
The Synthetic Fallback	Subdomain Takeover	7.5	0.20*	NO	Tier 2 (+0.5)	5.1	ATTEND

*Synthetic EPSS fallback applied for non-CVE risks. Negative floor logic ensures Tier 4 low-impact risks never drop below 0.0.

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The Live Risk Register

Unified Source:
25 columns fully auto-populated from all 14 sources. Zero manual entry required.

Live Risk Register

Dashboard
Risk
Assets
Reports
Settings

Risk ID	Description	Source	Status	Composite Score	Owner	Due Date	SLA
CVE-2024-29496	SharePoint Privilege Escalation	EASM	[CRITICAL]	10.0 WEAPONIZED	App Team	Apr 12, 2024	1 Day Overdue
CVE-2024-31072	Cleo File Transfer RCE	Threat Intel	[CRITICAL]	10.0 WEAPONIZED	App Team	Apr 13, 2024	1 Day Overdue
CVE-2024-32111	PHP CGI Argument Injection	EASM	[HIGH]	8.8	App Team	Apr 14, 2024	2 Days Left
CVE-2024-33543	Microsoft Office HTML RCE	CrowdStrike	[HIGH]	6.3	Infra Team	Apr 15, 2024	3 Days Left
CVE-2024-34765	Apache Struts Vulnerability	Threat Intel	[REMEDIATED]	0.0	SecOps	Apr 10, 2024	Closed

Intelligent Sorting:
Sorted dynamically by Composite Score—guaranteeing teams work on the worst, actively weaponised threats first.

Live Status: Every finding is scored, enriched, assigned to a named owner, and tracked against an SLA countdown timer.

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Threat Intelligence Overlay

Contextual intelligence applied on top of every risk row to answer:
Who is attacking, how, and are they targeting your specific sector?

Geo Threat Map

Live visualisation of global attack clusters.

APT Overlays

Highlighting specific actor groups (e.g., APT28, Lazarus, Sandworm) mapped directly to active CVEs in your environment.

APT28 (RU): CVE-2024-3400 (PAN-OS) - HIGH
Lazarus (KP): CVE-2023-20198 (Cisco IOS) - HIGH
Sandworm (RU): CVE-2024-31497 (PuTTY) - MEDIUM

Sector Focus

Filtering intelligence exclusively to threats actively targeting the enterprise's industry vertical.

Industry Vertical: Financial Services
Targeted Threats: 14 Active Pulses
Top Actor: APT33 (IR) - Energy/Finance Focus

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The Capability Gap: Excel vs. URIP Intelligence

	What You Have	Intelligence Added
Base Data	Excel (CVSS only = wrong priorities)	URIP (Composite engine)
Probability	Excel (Blind to likelihood)	URIP (EPSS Layer 2: 30-day exploit prediction)
Active Exploits	Excel (Static data)	URIP (KEV Layer 3: Confirmed weaponisation boost)
Business Impact	Excel (Every server treated equally)	URIP (Asset Tier Layer 4: Industry-standard RBVM context)
Attacker Context	Excel (No actor data)	URIP (MITRE/OTX Layer 5: Who, where, and how)
Decision Framework	Excel (Guesswork)	URIP (CISA SSVC Labels: Act Now / Act / Attend / Track)

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Risk Detail Exploded View

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From Insight to Execution

Phase 1: Intelligent Assignment

Auto-Routing

Findings map dynamically to specific domain owners (Network, Cloud, Identity).

SLA Timers

A Critical risk (Composite 9.0+) instantly triggers a 72-hour countdown and auto-pages the CISO.

Attached Steps

Fixed-text remediation steps (e.g., 'Add DMARC record to DNS') automatically appended to the ticket.

Phase 2: Auto-Remediation Execution

- Automated script library via CrowdStrike Real-Time Response (RTR) and Ansible.
- System pushes patches to endpoints or updates firewall rules without human intervention.

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The Auto-Remediation Engine

Execution Hubs

CrowdStrike RTR

Real-time response patching endpoints remotely.

Ansible Playbooks

Server configuration fixes and firewall rule adjustments.

CyberArk API

Auto-resetting compromised admin credentials.

Safety Callout

The ‘Implication Check’ ensures expected downtime and rollback plans are verified before any script executes.

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The Safety of Automation: The Execution Tree

1. Trigger: Weaponised risk detected & patch available via NVD API.

2. Implication Check (The Guardrail)
Before any script runs. URIP simulates the impact. What services will restart? What is the expected downtime? What is the rollback plan?

3. Approval Gate: Configurable by the CISO.

- Low-impact assets auto-execute
- Tier 1 assets trigger a "Require SPOC Approval" button.

4. Execution & Logging
Script pushed. Before/after state permanently logged.

5. Re-Test Loop
Asset automatically re-scanned. If fixed, risk closes automatically.

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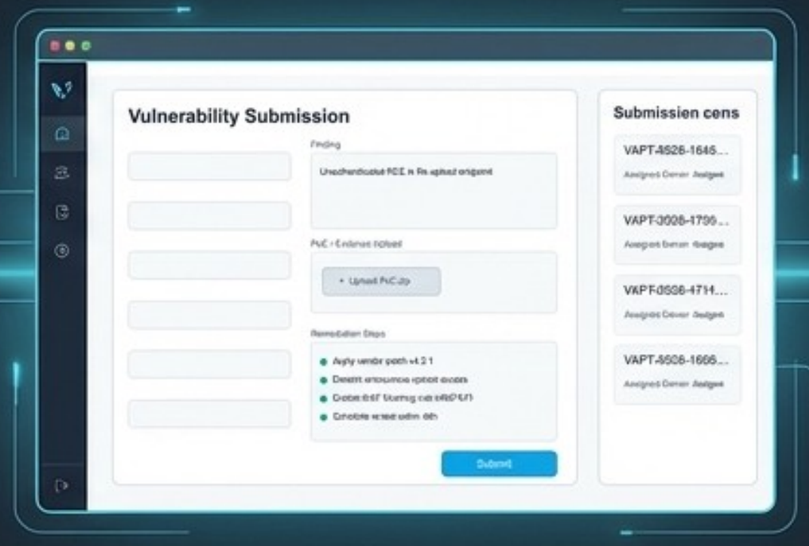
VAPT Vendor Portal: Eradicating Manual Parsing

Legacy Workflow

Vendor emails a 60-page PDF –> Internal team manually parses –> Enters into Excel –> Formatting errors & delays.

URIP Collaboration Module



- 1. Scoped Vendor Login: External pentester logs in securely to a sandboxed environment.
- 2. Structured Entry: Vendor inputs specific fields (CVE ID, CVSS, PoC upload) into a real-time validated form.
- 3. Auto-Processing: Instant submission triggers URIP enrichment (EPSS/KEV) and auto-assigns to the internal SPOC. Zero manual work.

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Role-Based Governance & Visibility

Role	Can See	Cannot See
CISO / Admin	✓ All 14 sources, full risk register, unmitigated vs. accepted risks.	✗ (Unrestricted).
IT Team / SPOC	✓ Only risks strictly assigned to their domain. SLA countdowns.	✗ Other department queues, external vendor details.
VAPT Vendor (Guest)	✓ Only their own submitted findings and re-test requests.	✗ Internal enterprise network data.
Executive / Board	✓ High-level KPI dashboards, trend charts, overall posture.	✗ Raw finding details or individual exploit strings.

Note:
Risk Acceptance Workflow requires digital sign-off from Head of Department. Accepted risks are automatically excluded from Board dashboards to prevent artificial inflation of risk posture.

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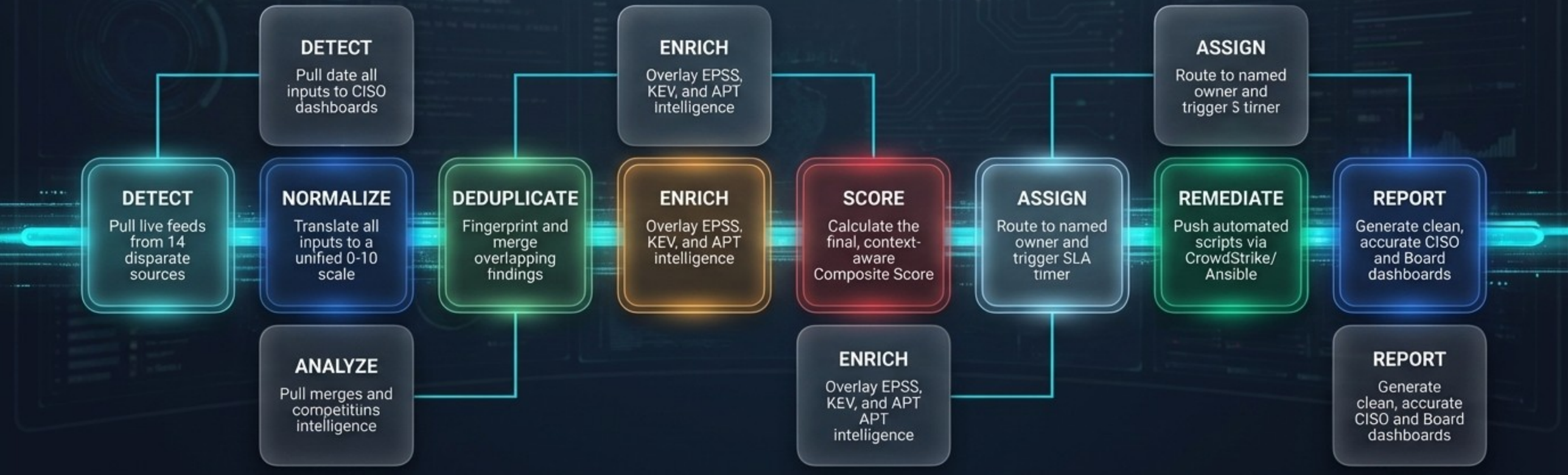
End-to-End Workflow: The Unified Orchestration Engine



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The Future of Orchestration: Total Control

- From **14 disconnected consoles** to **One Unified Schema**.
- From theoretical base scoring to **Weaponisation-First Prioritisation**.
- From **static spreadsheets** to **Automated SLA Enforcement**.
- From manual IT research to **One-Click Remediation**.

Stop sorting noise. Start acting on the signal.

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